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Objective and subjective assessment of sound diffuseness in musical venues via computer simulations and a scale model

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ABSTRACT

This study divides the effects of diffusion on the sound in an auditorium into objective and subjective aspects for assessment and proposes a new subjective assessment indicator, called "diffuseness." First, computer simulations of 12 performance halls having various shapes were carried out to examine the changes in the audience parameters with increasing diffusion coefficients of the walls and ceilings. The results indicate that most of the changes in the acoustic parameters, including the reverberation time (RT) and early decay time (EDT), were less than the just noticeable difference (JND), and there was no obvious tendency. However, when a sidewall diffuser was installed in a 1:10 scale model of a venue, the RT and strength (G) decreased, while the clarity (C_{80}) increased, owing to the increased sound absorption. The change was larger than that in the simulation and the initial reflection was found to increase. Nine subjective characteristics, including diffuseness, were evaluated for an audible sound source generated based on the binaural impulse response recorded at five points in the auditorium. The results show that despite the small changes in objective indicators, the attached diffuser made a clear and positive contribution to the overall impression as well as the subjective characteristics associated with diffuseness within the venue, such as intimacy and envelopment.

1. Introduction

Diffusive design is an important design element for improving the acoustic quality of a venue, and it has been found to influence the spaciousness of venues as well as the uniform distribution of acoustic parameters [1,2]. In addition, diffusion has been comprehensively shown to overcome acoustic defects, such as acoustic glare and tone coloration, due to strong specular reflection [3,4]. However, diffusion leads to negative side effects, such as source presence and proximity reduction, which reduce the uniqueness of the performance hall [5,6]. Thus, there is still some room for debate regarding the effects of diffusion. Given this background, to maximize the positive effects of diffusion, it is important to determine the location and shape of the diffuser during the design stage of a venue. Previous studies have used the sidewalls of the venue as the most effective sites to achieve diffuse reflection [7–9].

Over more than 30 years of research on the effects of acoustic diffusion on the acoustics in a venue, various aspects of diffusion, such as design, evaluation, prediction, and quantification, have been studied [10–13]. Initially, the random incidence scattering coefficient (Sc) and directional diffusion coefficient (Dc) were proposed to quantify the surface properties for diffusion. Subsequently, they were standardized in ISO 17497–1 and 17497–2 [14,15]. Scattering coefficients are obtained from specular reflection and are calculated as the proportion of the scattered energy to the total energy. Diffusion coefficients are used to represent the uniformity of the directional scattering distribution. Furthermore, methods for calculating the scattering coefficients using decay curves in non-diffusive rather than diffusive fields have been investigated [13]. However, as it is difficult to clarify the phenomenon of diffusion with only two indicators, various new indicators have been proposed to quantify diffusion [16–21]. Davy et al. [16] proposed the standard deviation of the reverberation time (RT) as an indicator of the diffuse sound field. The initial time delay gap and interaural cross correlation (IACC) were confirmed to change, owing to the diffuse sound field formed by the installation of a cylindrical diffuser. In particular, IACC was found to decrease with diffuser installation, confirming that it can be used as an indicator of the diffuse sound field [17,18]. In a recent study, Hanyu [19] proposed the scatter-to-absorption ratio, mean scatter time, and diffusion time as indicators of the diffuse sound field. Jeon et al. [20] proposed the relative standard deviation (RSD) obtained by

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dividing the mean value of each parameter by the standard deviation, to represent the variation between seats in the diffuse sound field after installation of the diffuser. A subsequent study [21] proposed the number of peaks (Np) as an indicator by quantifying the number of peaks that increase with the diffuse reflection produced by the diffuser. Consequently, the assessment of actual venues confirmed that RSD decreases and Np increases when diffusive design is applied. However, although most studies have proposed new meaningful parameters, they are limited in that they have not examined the relationships of these parameters with actual human subjective responses to diffusion.

For most venues, two methods are used to predict the objective sound in the early stage of design before implementation: scale modeling and computer simulation. With regard to the studies on diffusion using scale modeling, Rye and Jeon [22] found that the RT, early decay time (EDT), and clarity (C_{80}) decreased when hemispherical diffusers were installed in a reduced model. A subsequent study [23] examined the diffusion effects in 1:50 scale models of venues with different shoebox and fan-type shapes. It was found that after diffuser installation, RT and EDT decreased, while C_{80} generally increased. Chiles [24] found that EDT increased, and RT decreased after diffuser installation in a 1:25 scale model. Jeon et al. [25] found that diffusive design decreased RT, EDT, and G, while it increased C_{80}.

With regard to the studies on diffusion in a venue using computer simulation, initially, Cox et al. [13] found that it is appropriate to employ the scattering coefficient as an initial set value to predict diffusion in a room with acoustic prediction software. Using three simulation tools (CATT-Acoustic, Raven, and Odeon) with different diffusion algorithms, Shtrepi [26] found that EDT increased and C_{80} decreased when the diffusion coefficients of the inner walls and ceilings increased from 0.1 to 0.9. Subsequent studies [27,28] compared the method of implementing the diffuse sound field in a venue by changing only the diffusion coefficient of the flat walls with the method of realizing the diffuser module. The latter method was found to be more similar to reality in audible terms, and the source-to-receiver distance was found to be a key factor influencing diffusion. However, as discussed above, different assessment methods analyze the effects of diffusive design on the sound field of a venue differently. Therefore, it is necessary to examine the diffusion effect by combining various assessment methods.

In addition to verifying the objective effects on the diffusive design of a venue, studies have also been conducted to examine auditory effects. Takahashi [29] found that for periodic-type diffusers, the subjective tonal effects of responses increased with decreasing distance from the diffuser. In a subjective assessment of the diffusion of an audible sound source in a 1:10 scale model, the preference for the diffuse sound field was found to be strongly related to the loudness and reverberance [22]. In addition, through a subjective assessment based on simulation, the changes in reverberation, sound level, coloration, spaciousness, and source location after installing a diffuser were investigated [28], but no clear correlation was found with the objective parameters, suggesting that a new single measurable parameter is required to evaluate diffusion.

As mentioned above, although various studies have investigated diffusion in a venue, the objective or subjective effects of diffusion have varied across these studies and they have not been generalized. This may be because most previous studies have surveyed only a single venue, or they have selectively used a single methodology for simulation, scale modeling, or actual field measurement; thus, a comparative assessment of these studies is difficult. In addition, thus far, the subjective indicators of diffusion have not been clarified. Only the differences before and after the implementation of diffusive design have been compared, or the preferences in terms of reverberance or sound pressure have been evaluated. Therefore, there is a need not only for a new indicator of the subjective perceptive effects of diffusion itself but also for verifying its validity by reviewing its relationship with objective parameters.

Hence, based on previous studies, this study divided the effects of diffusive design on the acoustics of an auditorium into objective and subjective aspects, which were investigated via computer simulation and scale modeling, and a new diffusion assessment indicator was identified. First, the change in the sound field of a venue owing to diffusion was examined through 12 computer simulations. A 1:10 scale model was developed for a venue, and subjective assessments of audible sound sources were conducted on the basis of the binaural room impulse responses recorded at five locations to investigate the different perceptive responses before and after installing a diffuser. Finally, the validity of the proposed approach was verified by examining the relationship between the subjective indicators, including the newly proposed subjective diffusion assessment indicator, and the objective indicators.

2. Experiment I: objective assessment using 12 computer simulation ModelsV

2.1. Method

2.1.1. Selection of musical venues

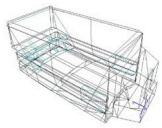
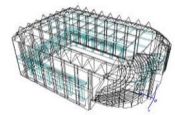
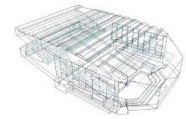
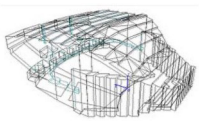
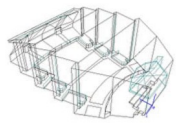
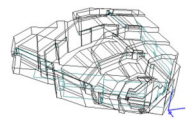
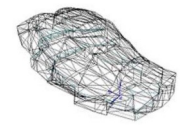
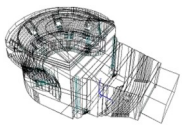
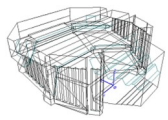
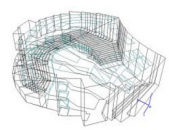
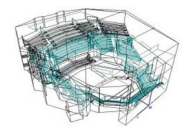
To investigate the effect of diffusive design on the sound in an auditorium, computer simulations were performed considering actual venues of various shapes, the details of which are presented in Table 1. In this study, the Odeon version 15 software was used for computer simulation. The 12 halls used in the simulation included four shoebox halls, four fan-type halls, and four other types of halls, all of which had 1,000 or more seats. Among them, Art Center Incheon (ACI), a performance hall in Incheon, Korea, was newly modeled. For the other 11 halls, the models included in the Odeon software library were used.

The basic reference values were used as they were, for the internal finishing materials of the 11 halls, embedded in the library. For ACI, the actual surface material of the hall was used. The library models use the sound absorption coefficient of the surface based on the data of an unoccupied concert hall according to ISO 3382 [30]. However, as RT is typically affected by the sound absorption effect of the seats, it was calculated under the occupied condition [3,9]; the coefficient of the surface material was adjusted to make it similar to the measured data of the seats occupied by the audience. When simulating the occupied concert hall, the sound absorption coefficient of a highly, moderately, or lightly upholstered occupied chair was selected for the chairs, as suggested by Beranek and Hidaka [31] for each hall.

2.1.2. Simulations

Odeon version 15 is a geometrical acoustic (GA) software program that uses a hybrid calculation method combining the image source method (ISM) and early scattering method for early reflections and the ray-tracing method for late reflections [32]. To implement a diffuse sound field, Odeon uses a vector-based scattering model that is based on linear interpolation for specular and diffuse reflection. In this case, the scattered vector is more realistic because it follows a random direction generated according to the Lambert distribution. The Odeon simulation considers both early and late reflections for scattered sound calculations. First, the early reflections reflect the secondary sources generated on each surface according to the transition order (i.e., change in the reflection order associated with a change from the early-ISM to the late-radiosity method) in addition to the image sources. The sound power is determined by the image sources and secondary sources distributed according to the scattering coefficients. For the late reflections, when a late ray is reflected by the surface, a small secondary source, similar to the surface source, is created, which reflects the oblique Lambert directivity, considering the surface roughness and edge diffraction. In addition, two other widely used GA-based software programs are CATT-Acoustic [33] and Raven [34]. First, for the late reflection, CATT-Acoustic uses a cone-axis tracing method called the randomized tail-corrected cone-tracing algorithm. For the early reflection, it uses the image source model up to the secondary reflection; to reflect the primary diffuse reflection, the secondary source, which

Table 1
Data used for objective evaluation of 12 concert halls.

Hall	Symphony Hall Boston (BO)	Groβer Musikvereinssaal Vienna (VM)	Concertgebouw Amsterdam (AM)	Royal Festival Hall London (LF)
Model Shape				
	Shoebox	Shoebox	Shoebox	Shoebox
Seats	2,625	1,680	2,037	2,901
Receivers	54	33	39	59
V [m^3]	18,750	15,000	18,780	21,950
S [m^2]	1,523	1,118	1,285	2,145
Hall	**Festspielhaus Salzburg (SA)**	**Konserthus Gothenberg (GO)**	**Gasteig Munich (MU)**	**Art Center Incheon (ACI)**
Model Shape				
	Fan	Fan	Fan	Fan
Seats	2,158	1,286	2,387	1,750
Receivers	47	44	47	41
V [m^3]	15,500	11,900	29,737	18,500
S [m^2]	1,555	836	1,819	1,472
Hall	**Usher Hall Edinburgh (EB)**	**Barbican Concert Hall London (LB)**	**Liederhalle Beethovensaal Stuttgart (ST)**	**St. David's Hall Cardiff (CW)**
Model Shape				
	Other	Other	Other	Other
Seats	2,502	1,803	2,000	1,952
Receivers	51	41	122	58
V [m^3]	15,700	17,000	16,000	22,000
S [m^2]	1,369	1,445	1,533	1,420

follows Lambert's distribution law, is generated on each diffuse surface [35]. Raven combines a stochastic ray-tracing model with an ISM to implement the diffuse reflection for the early and late reflections. It combines the principles of ray tracing and radiosity (diffuse rain) for more advanced models [36]. When examining the acoustic parameter changes with increasing scattering coefficients using the above three simulation tools, a previous study [26] found that some significant differences occurred (above JND) in EDT only in the Odeon simulation. Only in Odeon did the subjects clearly perceive an audible difference when diffusion was added. Accordingly, this study assumed that the Odeon simulation is the most suitable tool for evaluating the diffuse sound field.

In the computer simulations, the transition order was set to the second order and the impulse response length was set to 5000 ms to calculate the room acoustic indicators. The number of used late rays was 15,000, which was higher than the recommended value, allowing for comparison with other models [37]. For setting the sound collection points, the hall seats were divided into 2 m × 2 m grids, and one point was set in each grid. For ST, an asymmetrical hall, the grids were extended evenly to cover all the seats. The sound collection points were set at a height of 1.2 m. Thus, at least one sound collection point could be set for every 25 seats (see Table 1). The sound source was set to a height of 1.5 m in the solo position (1 m from the stage front and 1 m to the right) for all 12 halls.

To examine the effect on the audience through simulation, two scattering coefficients of the surface material in each hall were compared, as shown in Table 2. The scattering coefficient value was changed without changing the sound absorption coefficient of the base surface material. Only the sound scattering coefficients of the walls and ceiling of the auditorium, except for the stage and balcony, were changed. For the models without diffusive design, the reflective surface was realized by adjusting the scattering coefficient of the walls and ceiling to 0.05. For the models with diffusive design, a scattering coefficient of 0.7 was applied to the area under the same condition (absorption coefficients, air temperature, and relative humidity). In this study, the scattering coefficients were assumed to be 0.05 (reflective) and 0.7 (diffusive) to investigate the change in the overall audience sound caused by the diffusive design. The scattering coefficient was applied based on the 707-Hz mid-frequency band, similar to the average of 500–1,000 Hz, according to the assumption in the Odeon simulation [32].

The diffusive power was calculated as the product of the scattering coefficient and area of the inner surface of the halls. The absorptive power was similarly calculated as the sum of the products of the area and sound absorption coefficients of the materials inside the hall (see Table 2). As the scattering coefficient increased from 0.05 to 0.7, the diffusive power increased significantly, and a sufficient diffuse sound field was realized. The diffusive surface ratio was defined as the ratio of the diffusive surface area to the total surface area, and the ratio of the area with the scattering coefficient to the total area was in the range of 18–80%.

2.1.3. Room acoustical parameters

The reverberation time (RT_{20}), early decay time (EDT), strength (G), clarity (C_{80}), and lateral energy fraction (LF_{E4}) were analyzed for the 12 computer simulation models with different scattering coefficients. In

Table 2
Details of acoustical characteristics of the computer simulation model based on different scattering coefficients.

		Total surface area [m^2]	Absorptive power	Diffusive surface area [m^2]	Diffusive power		Diffusive surface ratio
					Sc = 0.05	Sc = 0.7	
Shoebox	BO	5756.2	1403.00	4518.7	226	3086	0.79
	VM	5149.3	844.87	4142.9	207	2581	0.80
	AM	6334.5	1370.45	3418.0	292	2676	0.54
	LF	8257.1	2215.57	3157.6	413	3249	0.38
Fan	SA	5179.4	1269.26	3391.0	249	2452	0.65
	GO	4210.8	982.73	2298.2	198	2112	0.55
	MU	9544.8	2147.04	5834.7	473	3833	0.61
	ACI	5309.3	1324.96	3198.7	160	4540	0.60
Other	EB	6095.0	1934.26	2747.3	265	1880	0.45
	LB	6718.5	1459.46	4176.9	307	3187	0.62
	ST	7494.7	1717.15	3229.7	374	3201	0.43
	CW	11683.0	2903.11	2136.0	573	3017	0.18

accordance with previous studies [3,9], RT_{20} was calculated with occupied seats and the remaining parameters were calculated with unoccupied seats. The RT_{20}, EDT, and G_{mid} values were expressed as an average of 500 and 1,000 Hz according to the frequency range specified in ISO 3382. C_{80} was calculated as an average at 500, 1,000, and 2,000 Hz, and LF_{E4} ('E' indicates the early sound and '4' indicates the average of the LF values) was calculated as an average of 125, 250, 500, and 1,000 Hz, according to previous research [3,9].

2.2. Results

Table 3 compares the acoustical parameters of the reflective and diffusive surfaces on the walls and ceiling around the auditorium. The range (minimum - maximum) of the measurements for each parameter is provided in Appendix A. The mean refers to the mean value of all the receivers. The RSD was obtained by dividing the standard deviation by the mean to examine and compare the variation between seats. In this case, as C_{80} showed a negative value, RSD was excluded.

The just noticeable difference (JND) of the room acoustic parameters is 5% of the minimum value for RT and EDT, 1 dB for G and C_{80}, and 0.05 for LF [30]. When diffusion was applied, the average value of all the halls was less than the JND for all the parameters, and there was no significant difference in RSD.

Considering each sound parameter, when changing from a reflective surface to a diffusive surface, the RT difference of five halls (BO, SA, GO, LB, and ST) was greater than the JND. In all the venues except the fan-type ones, RT was found to be similar or lower than the JND when changing from a reflective to a diffusive surface. The EDT difference was greater than the JND in four halls (BO, SA, GO, and ACI), and as with RT, the EDT value was similar or lower after the application of diffusion. For both RT and EDT, the variation between seats showed different trends for each hall, with no clear tendency. For G, it tended to decrease in general, but the change was less than the JND, and the deviation between seats increased slightly. This may be due to the loss of sound energy, owing to the increase in the sound absorption effect with the scattering coefficient [38]. C_{80} showed an overall increase, but the

Table 3
Summary of room acoustical parameters according to different scattering coefficients for 12 concert hall models. (bold numbers indicate delta values higher than the JND).

Scattering coefficient			RT_{20} [s]			EDT [s]			G_{mid} [dB]			C_{80} [dB]			LF_{E4}		
			0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ
Shoebox	BO	*Mean*	2.42	1.99	**−0.43**	2.14	2.03	**−0.11**	4.7	4.3	−0.4	−1.13	−0.83	0.3	0.23	0.24	0.01
		RSD	0.1	0.05	−0.05	0.09	0.09	0	0.3	0.5	0.2	–	–	–	0.22	0.22	0
	VM	*Mean*	2.72	2.71	−0.01	2.65	2.69	0.04	6.0	5.8	−0.2	−0.57	−0.57	0	0.2	0.2	0
		RSD	0.02	0.02	0	0.06	0.05	−0.01	0.4	0.5	0.1	–	–	–	0.3	0.3	0
	AM	*Mean*	2.3	2.3	0	2.44	2.38	−0.06	5.8	5.6	−0.2	−0.37	−0.2	0.17	0.14	0.13	−0.01
		RSD	0.01	0.01	0	0.04	0.04	0	0.2	0.2	0	–	–	–	0.47	0.47	0
	LF	*Mean*	1.57	1.53	−0.04	1.43	1.43	0	2.5	2.3	−0.2	1.47	1.43	−0.04	0.34	0.18	**−0.16**
		RSD	0.02	0.02	0	0.15	0.16	0.01	1.0	1.1	0.1	–	–	–	0.51	0.36	−0.15
Fan	SA	*Mean*	1.83	1.93	**0.1**	1.97	1.8	**−0.17**	5.3	4.7	−0.6	−0.57	−0.03	0.54	0.2	0.17	−0.03
		RSD	0.02	0.02	0	0.11	0.13	0.02	0.3	0.4	0.1	–	–	–	0.44	0.37	−0.07
	GO	*Mean*	1.59	1.71	**0.12**	1.65	1.84	**0.19**	5.5	5.5	0	0.4	−0.43	−0.83	0.16	0.16	0
		RSD	0.02	0.03	0.01	0.06	0.04	−0.02	0.3	0.3	0	–	–	–	0.4	0.34	−0.06
	MU	*Mean*	2.25	2.23	−0.02	2.4	2.4	0	2.3	2.0	−0.3	−1.87	−2.07	−0.2	0.22	0.24	0.02
		RSD	0.05	0.03	−0.02	0.11	0.16	0.05	1.3	1.6	0.3	–	–	–	0.43	0.41	−0.02
	ACI	*Mean*	2.14	2.17	0.03	2.44	2.25	**−0.19**	5.8	6.0	0.2	−0.60	0.10	0.7	0.21	0.22	0.01
		RSD	0.05	0.06	0.01	0.1	0.12	0.02	0.4	0.4	0	–	–	–	0.32	0.25	−0.07
Other	EB	*Mean*	1.67	1.68	0.01	1.88	1.85	−0.03	4.0	4.1	0.1	−0.3	0.73	**1.03**	0.2	0.21	0.01
		RSD	0.03	0.02	−0.01	0.09	0.1	0.01	0.6	0.6	0	–	–	–	0.57	0.56	−0.01
	LB	*Mean*	2.13	1.99	**−0.14**	2.10	2.07	−0.03	3.7	3.5	−0.2	−1.73	−1.67	0.06	0.15	0.16	0.01
		RSD	0.02	0.02	0	0.05	0.04	−0.01	0.4	0.5	0.1	–	–	–	0.49	0.44	−0.05
	ST	*Mean*	1.97	1.84	**−0.13**	2.04	2	−0.04	3.8	3.6	−0.2	−0.60	−0.5	0.1	0.16	0.17	0.01
		RSD	0.04	0.07	0.03	0.08	0.07	−0.01	0.6	0.6	0	–	–	–	0.51	0.45	−0.06
	CW	*Mean*	2.12	2.12	0	2.01	2.04	0.03	2.3	2.3	0	−0.50	−0.17	0.33	0.18	0.2	0.02
		RSD	0.05	0.04	−0.01	0.1	0.1	0	1.2	1.4	0.2	–	–	–	0.48	0.54	0.06
Average		*Mean*	2.05	2.02	−0.03	2.10	2.06	−0.04	4.3	4.2	−0.1	−0.42	−0.23	0.19	0.20	0.19	−0.01
		RSD	0.04	0.03	−0.01	0.09	0.10	0.01	0.6	0.7	0.1	–	–	–	0.42	0.39	−0.03

change was less than the JND. In addition, LF_{E4} decreased overall in all halls, and exhibited a change greater than the JND only in the Royal festival hall London (LF). Furthermore, RSD was either similar or reduced. The reduced proportion of lateral energy causes the change in this value, as with G, because some of the strong early reflections from the walls and ceiling are lost, owing to the sound absorption effect due to increased diffusion in the walls and ceilings. As such, most changes in the objective acoustic indicators of the diffusive design in the concert hall showed small changes less than the JND, and at the same time, there was no obvious increasing or decreasing tendency. Thus, it is difficult to generalize the room acoustical changes due to diffusion. There are various causes of this phenomenon; one of these is that the performance of the GA software varies with the method (precision) used for modeling the hall structure [39,40]. However, the main reason is that generalization is difficult because the shape of the music venue itself is different from that considered in the algorithm [23,41]. Therefore, the simulation may be useful to predict the sound field for one hall in advance; however, if the purpose is to generalize changes in the diffuse sound field, then there are limitations for generalizing the changes in physical parameters; it is not possible to generalize changes in the diffuse sound field by simply varying the scattering coefficient without considering the shape or internal finishing characteristics of each hall.

3. Experiment II: subjective assessment using a scale model

3.1. Method

3.1.1. Scale model

To examine the subjective perceptive change in the sound before and after the application of diffusive design to the musical venue, a scale model was developed, and an audible experiment was conducted. First, one hall (ACI) was selected from the halls simulated in Experiment I, for which a 1:10 scale model was developed. The basic shape of the hall is the vineyard style, which fully exploits the fan-type design. The hall was designed for acoustic and visual intimacy, with a capacity of 1,750 seats (18,500 m^3 in volume). The average height per width (H/W) and average length per width (L/W) of the concert hall are 0.7 and 1.1, respectively.

Fig. 1 (a) shows the 1:10 scale model after diffusion. Sound-absorbing materials were used only for the auditorium and chorus seats, and the surface was made hard and smooth for other locations. To compare the diffusion characteristics of the concert hall with and without the diffuser, measurements were made before and after the diffuser was installed at the sidewalls and balcony.

As shown in Fig. 1 (b), a diffuser with a higher scattering coefficient was placed at the periphery of the seats near the stage. As the distance increased, a diffuser profile with a lower scattering coefficient was applied [23,42]. Three diffusers (see Fig. 2 (a)) were applied to the scale model. The placement of the diffusers was determined based on the scattering (Sc) and diffusion (Dc) coefficient measurement data determined through diffusion performance experiments using the 1:10 diffuser sample, as shown in Fig. 2 (b) and (c).

The profile of the diffuser was designed as a vertically projected shape based on a quadratic residue diffuser to spread the sound evenly and horizontally. As shown in Fig. 2 (a), the cross section of the diffuser was transformed into a combination of different depths of 10, 80, and 60 mm according to Type 0, 1, and 2, respectively. The scattering/diffusion coefficients were determined based on ISO 17497–1 [14] and ISO 17497–2 [15]. In the 1/3 octave band in the frequency range of 100–5000 Hz, the mean values were 0.04/0.19, 0.25/0.30, and 0.39/0.37, respectively. The sound absorption coefficient of the diffuser was also measured in the 1:10 scale model. As shown in Fig. 2 (d), the average sound absorption values in the octave band of the 125–4,000 Hz band were 0.14, 0.17, and 0.20, respectively. After attaching the diffuser, with a diffuser surface area of 264.8 m^2, the diffusive surface ratio was approximately 5% of the total surface area (5309.3 m^2). Here, the diffusive power increased by approximately 4.5 times after attaching the diffuser (2753), compared to that before attachment (616).

3.1.2. Stimuli

To create stimuli for subjective evaluation, as shown in Fig. 3 (a) and (b), a 1:10 scaled omnidirectional loudspeaker (dodecahedron loudspeaker) and a dummy head equipped with two 1/8-inch microphones (B&K) were used as the devices for the sound source playback and sound reception, respectively. The compact loudspeaker, consisting of a speaker unit having a diameter of 13 mm, was created using a 3D printer. The omni-directionality of the sound source has been demonstrated based on ISO 3382–1 [30]. The maximum and minimum directivity indices for different directions were found to meet the requirements in the range of 1–50 kHz. The upper part of the small dummy head, which includes the head and ears, was also created using a 3D printer to fit the standard build of a Korean male. The position of the sound source was set to the position of the solo player as in the simulation, i.e., 1 m (10 cm in the 1:10 scale) away from the central edge of the stage horizontally and vertically. As shown in Fig. 3 (c), five representative points (R1–R5) were selected, which were influenced by different diffusion planes along the sidewalls. The selected points (R1–R5) were adjacent points within 5 cm (actually 5 m) of the left sidewall in the direction from the stage to the seat, and their distances from the diffuser were 3.8, 2.5, 1.5, 2.5, and 5.0 cm, respectively. The played sound source was a sine sweep. The recorded sound sources for the left and right ears were inverse-sine swept and de-convoluted to produce a binaural room impulse response (BRIR). In this study, a sample rate of 192 kHz was used because the 1:10 scale model requires a frequency in the 10-fold range (100 Hz–80 kHz) according to the similarity law of sound (scale model). After measurement, the sample rate was down sampled to 19.2 kHz using the Nyquist frequency and *anti*--aliasing filtering with the Adobe Audition software. This study focused on the sound field changes before and after diffuser installation. Therefore, the air absorption for the impulse response was not considered. However, to design the boundary conditions before and after the diffuser installation as close as possible, the interior was sealed and

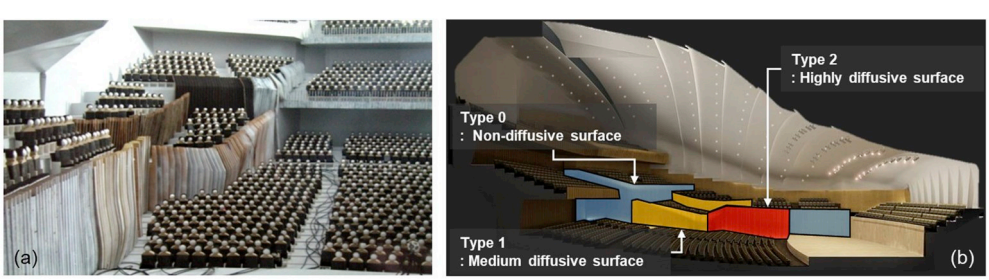


Fig. 1. Details of the scale model: (a) 1:10 scale model view and (b) location and type of three diffusers: Type 0, Type 1, and Type 2.

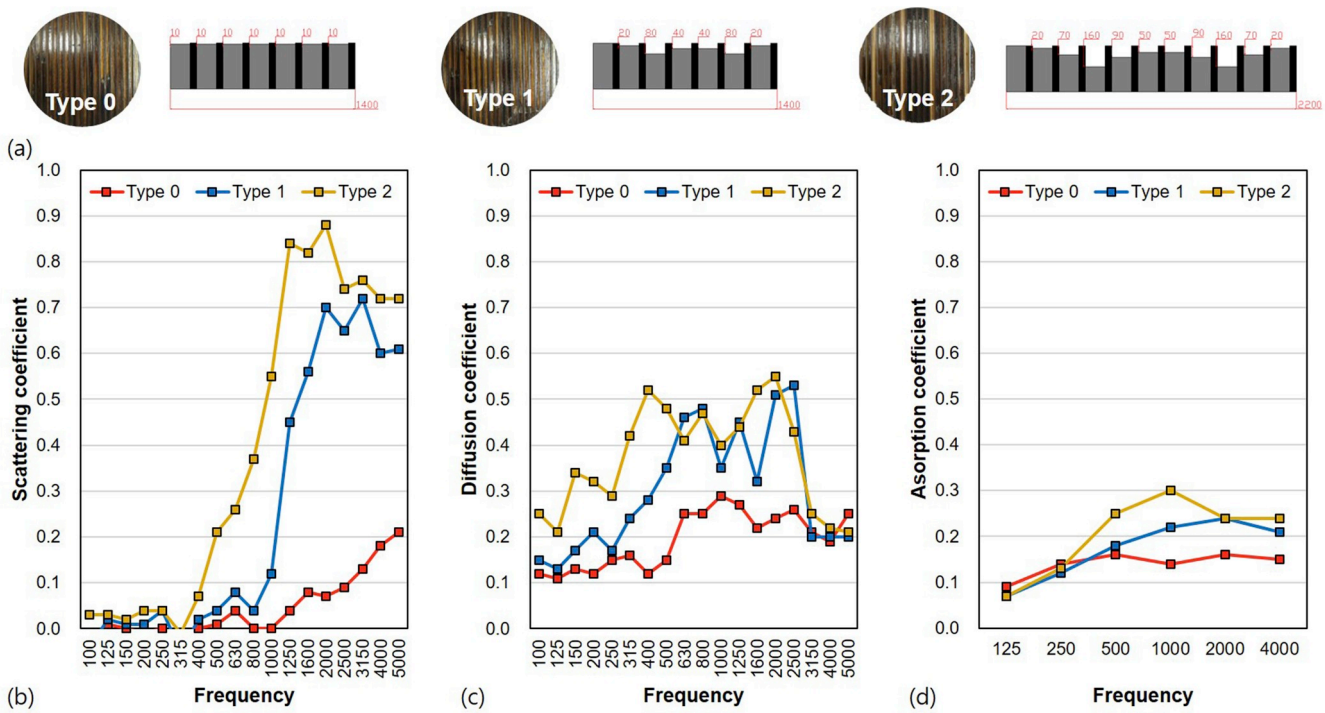


Fig. 2. Acoustic characteristics of three types of diffusers: (a) section view, (b) scattering coefficient, (c) diffusion coefficient, and (d) absorption coefficient.

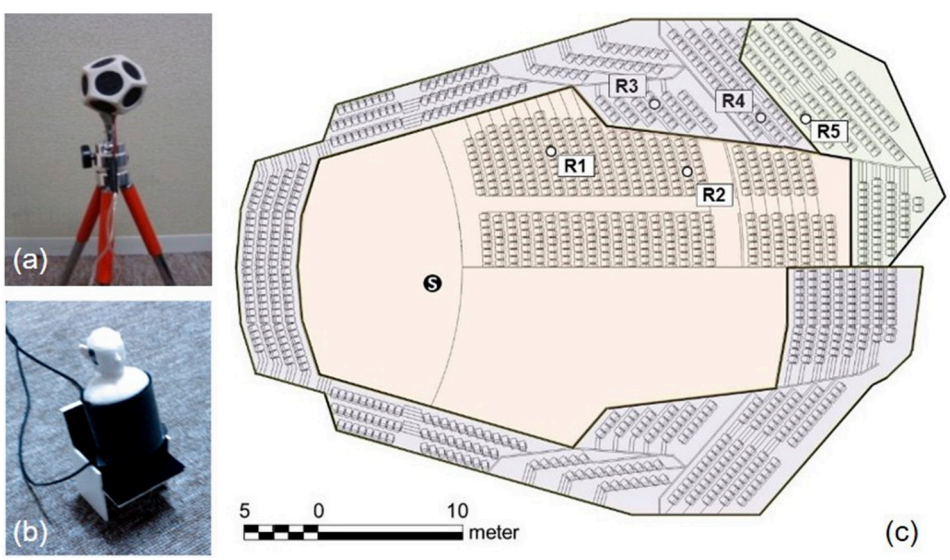


Fig. 3. Experimental setups: (a) 1:10 scale omnidirectional loudspeaker, (b) 1:10 scale dummy torso, and (c) locations of source (the black circle denotes a soloist position, S), receivers (white circles), and diffusers in a three-dimensional view of the concert hall.

charged with nitrogen to maintain constant humidity. To minimize the change in temperature, the measurements were taken in a short time.

For auditory evaluation, the soprano sound source measured in the anechoic chamber and BRIRs calculated at five different points, with and without the diffuser, in the scale model were convoluted. Such auralization technology can implement different sounds in terms of diffusion recognition three-dimensionally. The soprano sound source used in the evaluation was an extract from "Non t'amo più! – Francesco Paolo Tosti". It is a sound source with less harmonic overtones and for which the diffusion can be distinguished easily by the syllables. The length of the sound source was 15 s, and the pilot test was set based on the time when the test subjects responded that they could give a sufficient evaluation but were not bored. As shown in Fig. 4, when the spectral characteristics of the soprano sound source were examined, the source range was varied from a low frequency to a high frequency in the range of around 100 Hz to 4 kHz, which included the basic frequency range for evaluating the sound field of a concert hall. In fact, the soprano sound source has a strong directivity within the venue. As this study focused on the change in the sound field of the audience with and without the diffuser, the sound source was played omni-directionally.

For the stimuli, in a semi-anechoic room, each subject was provided with 10 sound sources before and after installing the diffuser once in a random order through an audio interface (Ronald Octa-capture UA100) and headphones (Sennheiser HD 650). The subjects were allowed to listen repeatedly if required. The background noise was 25 dBA.

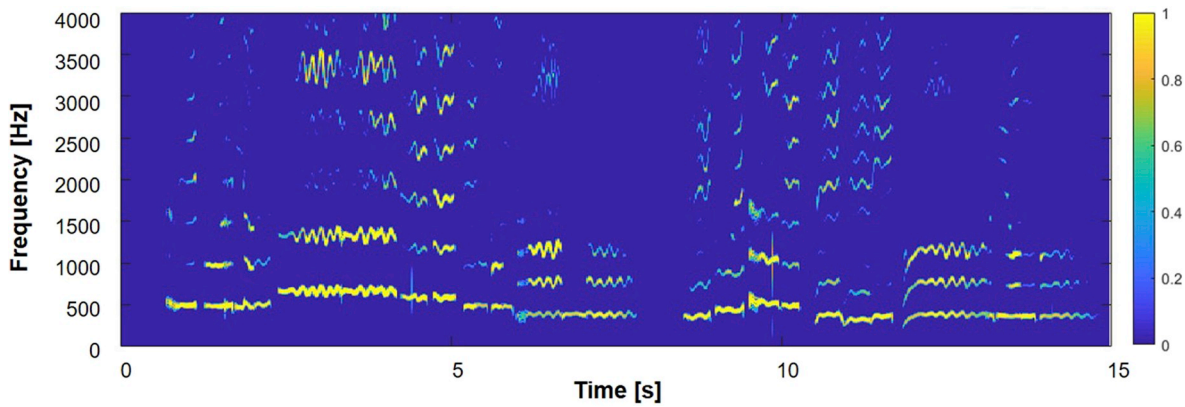


Fig. 4. Spectral content of the soprano sample used in the listening test.

3.1.3. Subjective characteristics

To examine the changes in the subjective responses before and after installing the diffuser in the venue, auditory evaluation was conducted by 30 audio engineers with considerable auditory experience as tonmeisters. The subjects were 20–45 years of age. Prior to evaluation, an audiometer (Rion AA-77) was used to assess the subjects' hearing limits. They were all found to have normal hearing. For subjective evaluation of diffusion, nine subjective characteristics (clarity, reverberance, envelopment, intimacy, loudness, brilliance, warmth, diffuseness, and overall impression) were defined (see Table 4). In addition, it was determined that the comparison of individual values for the four diffuseness items could not indicate the response of the subjects to subtle differences in diffusion; therefore, the overall mean was expressed as the representative value of diffuseness. Accordingly, 12 items were evaluated on an 11-point scale.

Subjective properties 1–7 were classified as acoustic impressions of the interior space [43], properties 8–11 were classified as the perception of diffusion, and property 12 was classified as the overall acoustic satisfaction. Detailed explanations of all the items were provided before the experiment so that the subjects could fully understand them.

In particular, diffuseness was evaluated in relation to the temporal (8−10) and spatial characteristic (11). First, the "density of reflection" was used to examine the effect of the reflections added by diffusion among the specular reflections from the wall according to the diffusive design [21]. Diffusion may have softened the tone of the strong reflections on the sidewalls; however, a slightly reduced delay gap could likely affect the early-to-late sound index, owing to additional reflections. Therefore, the subjects were asked to focus on the early stage of the soprano sound generation. "Smoothness of decay" means the smoothness of the decay of the late reverberation softened by the diffusion. In other words, the subjects were asked to concentrate on the late reverberance and level change of the sound source. For "smoothness of reflection," the subjects were asked to focus on the positive effect of diffusion on reducing acoustic glare and identify whether the echoes and coarse specular reflections were softened. In other words, this item aimed to determine the possibility of resolving acoustic defects or audible problems from the generation of a sound source to its disappearance. "Isotropic directivity" was intended to determine whether the diffusion of negative energy was homogeneous in all directions. The subjects were free to express their perception regarding the width of the sound source and the homogeneity of the sound direction perceived binaurally.

Table 4
Description of subjective impressions.

Indicator	Descriptions	Rating	
		Lowest (0)	Highest (10)
1 Clarity	Sound is clear	Muddy	Clear
2 Reverberance	Amount of reverberation when a note or tone stops	Dead	Live
3 Envelopment	Sound surrounds me sufficiently	Constricted	Expansive
4 Intimacy	Subjective impression of listening to music in a large room and its sound even though the room is small	Remote	Intimate
5 Loudness	Sound is sufficiently loud	Quiet	Loud
6 Brilliance	Sound is rich in high-frequency tones	Low	Strong
7 Warmth	Sound is rich in bass tones	Low	Strong
8 Diffuseness			
Density of reflection	Temporal density of reflections	Sparse	Dense
Smoothness of decay	Reverberation decay is smooth	Uneven	Even
Smoothness of reflection	Sound itself or reflection is smooth	Specular	Smooth
Isotropic directivity	Directivity of sound is homogeneous from all directions	Non-homogeneous	Homogeneous
9 Overall impression	Overall impression and preference for the sound	Very poor	Excellent

3.1.4. Normalization

When the subjects assess their subjective perception of diffusion, there is a possibility that, despite using the same scale, they do not perform the evaluation on the basis of the same criteria, owing to individual differences (expected values, interpretation differences, sensitivity, etc.). Therefore, it is necessary to normalize the range of the subjective evaluation responses before averaging the subjective response values. The variance of each subject's response was normalized according to Equation (1), which has been used in previous studies [43, 44], and the variance was then converted into a constant.

$$X_{norm,i,j,p} = X_{i,j,}\sqrt{\frac{\overline{\sum_{i,j} X_{i,j}^2}}{\sum_{i,j} X_{i,j,p}^2}}, \tag{1}$$

where i denotes the stimuli, j denotes the questions, $X_{i,j,p}$ denotes the initial answer of person p for stimulus i and question j, $X_{norm,i,j,p}$ denotes the normalized answer of person p for stimulus i and question j, $\sum_{i,j} X_{i,j,p}^2$ denotes the sum of squares of all the answers for person p, $\overline{\sum_{i,j} X_{i,j}^2}$ denotes the average of the sum of squares for all the subjects, and $\overline{\sum_{i,j} X_{i,j}^2} = \sum_{p=1}^{n}\left(\frac{1}{p}\right)\sum_{i,j} X_{i,j,p}^2$.

3.2. Results

3.2.1. Acoustical parameters

Before considering the subjective response to diffusion, the acoustical parameters were analyzed to examine the changes in the sound field before and after applying the diffusive design to the hall model. The results are presented in Table 5. According to ISO 3382–1 and previous research [3,9], acoustic parameters, such as RT_{20}, EDT, C_{80}, G_{mid}, and 1-$IACC_{E3}$, were examined. The audience conditions were expressed using the acoustical parameters, which were set to the same values as in the simulation. The brilliance, base ratio, and number of peaks (Np) were analyzed along with them, where the G value was calculated on the basis of the sound pressure level of the sound recorded at a distance of 10 m in the anechoic chamber, and it was expressed as an average of 500 and 1,000 Hz. The binaural quality index (BQI = 1-$IACC_{E3}$) refers to the difference in musical sounds between the ears within 80 ms. The interaural cross correlation ($IACC_{E3}$) was analyzed for the initial 80 ms of the direct sound in BRIR and it was calculated as an average of 500, 1,000, and 2,000 Hz. The brilliance was evaluated as the ratio ($RT_{2000} + RT_{4000}$)/($RT_{500} + RT_{1000}$), and the bass ratio was calculated as ($RT_{500} + RT_{1000}$)/($RT_{125} + RT_{250}$). These indicators are related to the harmonic richness, bass richness, and timbre, respectively. In this study, Np, proposed as a quantitative indicator of diffusion in a previous study [21], was analyzed as well. It denotes the number of peaks that exist within the 20-dB attenuation time of the direct sound in the impulse response time domain. Of course, as Np is not an absolute value, it has a limitation in that it cannot be compared in different spaces. However, the relative diffusion at each point can be compared; hence, it was analyzed in the same scale model.

Examining the changes in acoustic parameters before and after diffuser attachment indicated that RT, EDT, G_{mid}, and bass ratio decreased overall, while C_{80}, 1-$IACC_{E3}$, brilliance, and Np increased. In addition, the standard deviations of the indices related to early reflection (EDT and 1-$IACC_{E3}$) increased, and the standard deviations of all the remaining acoustic parameters decreased. In particular, the standard deviation of the bass ratio significantly decreased by 0.18. On average, RT_{20} and G_{mid} decreased by 0.17 s and 0.4 dB, respectively, in all seats, and the standard deviation decreased by 0.03. EDT showed different trends according to the distance from the sound source; it increased in the seats close to the sound source (R1-R3) and decreased in the rear rows. Similar to EDT, after attaching the diffuser, C_{80} increased in rows further to the rear (R3-R5). 1-$IACC_{E3}$ also increased in rows further to the rear. Brilliance and Np increased in all seats, and the bass ratio decreased in most seats after applying the diffuser.

Compared to the JND standard of each parameter, RT significantly decreased (JND = 5%) after diffuser attachment at all points except R2, and EDT (JND = 5%) increased or decreased depending on the point, though the variations were much larger than the JND. C_{80} (JND = 1 dB) exhibited significant differences only at R1 and R4, and G was less than the JND (1 dB) in all seats. These changes in the parameter values can be attributed to the considerably dominant energy reduction during the early specular reflection owing to the sound absorption effect from the attached diffuser. Certain parameters may have been affected by the introduction of early diffuse reflections generated by the diffuser profile, considering that the EDT at the points close to the sound source (R1–R3) increased after diffuser application.

Considering each acoustic parameter more closely, first, RT decreased at all the points as the diffuser was attached to the wall. This is because the overall sound absorption increased as the diffuser occupied 20% of the sidewall area [45,46]. Despite the sound absorption effect of the diffuser, EDT increased at R1–R3. As shown in Fig. 5, the decay pattern caused by the diffuse sound showed a gentle curve up to around 300 ms of the early decay, which resulted in the increased EDT. However, the late reverberation was attenuated at all the points, resulting in a short RT. As the intensity of the reflection is dependent on the source-to-receiver distance, EDT was reduced in the rear seats (R4, R5) with many late reflections due to the reduction in the initial reflection itself rather than the diffuser effect. Therefore, as seen thus far, the audience around the stage, which is affected by the diffuser, and the rear seats, which are affected by the late reflections, show a clear difference in the impulse response and decay curve. Therefore, referring to the results of the auditory evaluation presented in Sec. 3.2.2, if diffusive design is applied to the space to change EDT by a level above the JND, the difference between the diffuse and reflected sound fields can be identified perceptually. In this case, the increased reflection density through the diffuse reflection will affect the early decay curve, and the sound absorption of the diffuser itself will affect the amplitude of the late reflection. The brilliance increased by an average of 0.1 (14%) in all the seats, and the bass ratio decreased to 0.3 (20%) on average. The frequency characteristic of RT shown in the changes in the brilliance and the bass ratio is a common phenomenon due to the sound absorption coefficient of the pane-type diffuser when the diffuser is installed at an

Table 5
Comparison of acoustic parameters measured in diffusive and non-diffusive cases. (bold numbers indicate delta values higher than the JND for RT_{20}, EDT, C_{80}, G_{mid}).

	Diffuser	RT_{20}[s]	EDT[s]	C_{80}[dB]	G_{mid}[dB]	1-$IACC_{E3}$	Brilliance	Bass ratio	Np
R1	Without	2.25	2.3	0	7.1	0.71	0.76	1.2	72
	With	1.99	2.46	−2.3	6.6	0.7	0.83	1.2	88
	Δ	**−0.26**	**0.16**	**−2.3**	−0.5	−0.01	0.07	0	16
R2	Without	2.1	2.05	−1.6	6.4	0.6	0.62	1.3	73
	With	2.05	2.16	−2.1	5.9	0.55	0.78	1.3	102
	Δ	−0.05	**0.11**	−0.5	−0.5	−0.05	0.16	0	29
R3	Without	2.22	2.11	−0.9	5.8	0.71	0.75	1.8	86
	With	1.98	2.33	−0.5	5.4	0.71	0.77	1.2	95
	Δ	**−0.24**	**0.22**	0.4	−0.4	0	0.02	−0.6	9
R4	Without	2.19	2.33	−2.5	3.9	0.66	0.72	1.5	70
	With	2.02	1.82	−0.8	3.6	0.74	0.85	1.3	92
	Δ	**−0.17**	**−0.51**	**1.7**	−0.3	0.08	0.13	−0.2	22
R5	Without	2.13	2.41	−2.2	3.8	0.56	0.7	1.5	96
	With	2.01	2.24	−1.9	3.4	0.78	0.81	1.2	107
	Δ	**−0.12**	**−0.17**	0.3	−0.4	0.22	0.11	−0.3	11
Ave	Without	2.18	2.24	−1.4	5.4	0.65	0.71	1.5	79
	With	2.01	2.2	−1	5	0.7	0.81	1.2	97
	Δ	−0.17	−0.04	0.4	−0.4	0.05	0.1	−0.3	18
Stdev	Without	0.06	0.15	1.01	1.49	0.07	0.06	0.23	11.22
	With	0.03	0.24	0.81	1.42	0.09	0.03	0.05	7.66
	Δ	−0.03	0.09	−0.2	−0.07	0.02	−0.03	−0.18	−3.56

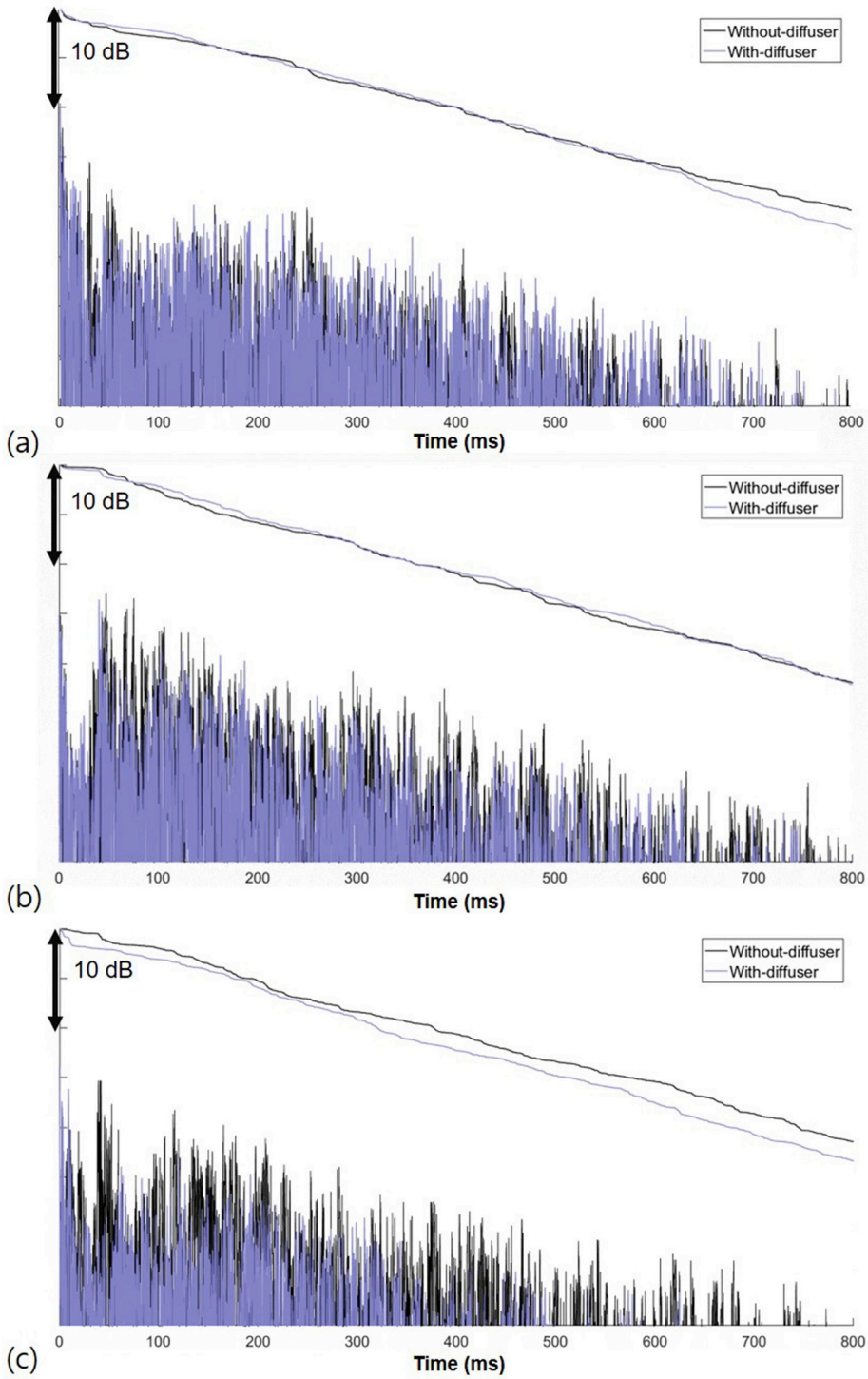


Fig. 5. Comparison of impulse responses and Schröder decay curves for diffusive and non-diffusive cases at three positions: (a) R1, (b) R2, and (c) R3.

effective level. Np tended to increase owing to the diffuse reflection generated upon applying the diffuser.

3.2.2. *Perceived diffuseness*

Diffuse reflections that occur after the direct sound cause differences in the subject's subjective responses as well as changes in the objective acoustic indicators. The results of the nine subjective indicators discussed in this study are shown in Fig. 6. The mean rating scores of all the subjective characteristics, except reverberance and warmth, increased after applying the diffuser. The mean difference between the subjective indicators before and after diffuser installation was statistically significant ($p < 0.01$). The mean score ratings of reverberance are higher than 5, which corresponds to the middle of the 11-point scale, for both cases. It can be inferred from this that the reverberant sound field was achieved in both cases. The overall impression of the musical venue was improved after installation of the diffuser, and the envelopment of the space increased, confirming the positive effect of diffusion. As can be seen in Table 5, although the sound strength, a physical indicator, which was less than the JND, tended to decrease, subjective indicators such as loudness, intimacy, and clarity increased. This may be due to the

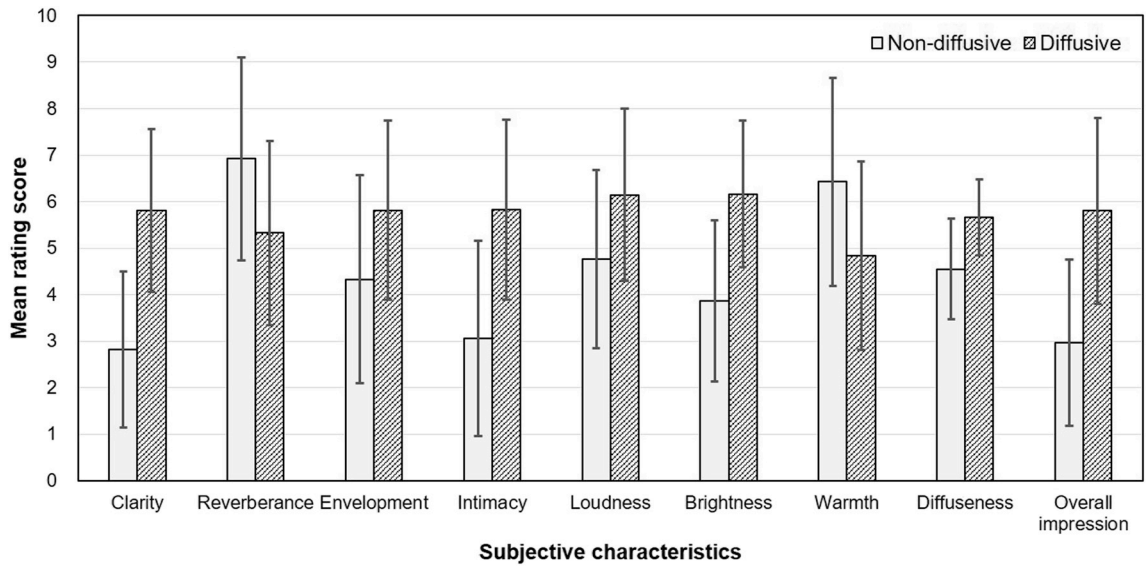


Fig. 6. Subjective impressions in diffusive and non-diffusive cases. The error bars represent the standard deviation.

increased density of early reflections produced by diffuse reflection, as can be inferred from the tendency of EDT, C_{80}, and Np after diffusion. The diffuseness represents the mean values of the density of reflection, smoothness of decay, smoothness of reflection, and isotropic directivity. The diffuse sound field generated by the diffuser seems to be perceived clearly to some extent. Moreover, the variation in the four subjective evaluation factors for diffuseness was approximately 1.11, relatively small compared to the other evaluation indicators. However, this showed a statistically significant increase after diffuser installation ($t(149) = -10.012$, $p < 0.01$). Additionally, the magnitude (d) of the effect indicating actual significance was also very high, at 0.82 (great effect). Table 6 presents the results of the Pearson correlation analysis performed to examine the linear relationship between the subjective response indicators. Reverberance and warmth showed a negative correlation with the other subjective indicators. In particular, overall impression showed a high correlation with clarity, reverberance, intimacy, and warmth, with correlation coefficients of 0.58, −0.48, 0.58, and −0.42, respectively ($p < 0.01$); further, it showed a positive correlation with diffuseness with a correlation coefficient of 0.42, implying a positive correlation at a level similar to that of the other subjective indicators ($p < 0.01$). The subjective perception of diffuseness shows a high negative relationship with a correlation coefficient of −0.42 with reverberance and warmth, indicating that the sound absorption effect due to diffuser installation was very high.

Table 6
Pearson's correlation coefficients between subjective impressions. Only cases with significance, $p < 0.01$, are presented (Re: Reverberance, En: Envelopment, In: Intimacy, Lo: Loudness, Br: Brightness, Wa: Warmth, Di: Diffuseness, Ov: Overall impression).

	Re	En	In	Lo	Br	Wa	Di	Ov
Clarity	−.47	.21	.62	.30	.53	−.39	.58	.34
Reverberance		−.25	−.55	−.22	−.34	.33	−.48	−.42
Envelopment			.28	.27	.35	−.41	.25	.19
Intimacy				.35	.48	−.43	.58	.31
Loudness					.45	−.29	.20	.17
Brightness						−.43	.38	.32
Warmth							−.43	−.42
Diffuseness								.42

A dependent *t*-test was performed on the average difference of the subjective characteristics with and without diffuser installation at each evaluation point. The results are shown in Fig. 7. Considering each seat, R1–R4 showed statistically significant differences in the clarity, intimacy, loudness, brightness, warmth, diffuseness, and overall impression after installation of the diffuser. The magnitude of the effect was also greater than a moderate effect. R5, located at the farthest source-to-receiver distance, showed a statistically significant increase in the clarity, diffuseness, and overall impression; however, overall, the differences in the subjective responses due to diffusion were not as significant. For R5, the diffuser was not applied to the nearby wall, and the source-to-receiver distance was extremely large. Therefore, the direct effect of the diffuser may have been moderately smaller than that at the other points.

3.2.3. *Relationship between objective parameters and subjective attributes*

To examine the relationship between the objective acoustic parameters and subjective attributes, the Pearson correlations presented in Table 7 were obtained. It is evident that the subjective responses form the strongest relationship with RT among the various acoustic parameters. This can be attributed to two influencing factors: 1) increase in sound absorption due to the increase in the surface area inside the hall owing to the diffuser, and 2) changes in perception of reverberance and coloration according to the diffusion effect [26,28] due to diffuser application. This can also be inferred through significant relationships among other related reverberance indicators as well (bass ratio or brilliance). Furthermore, when considering both diffuseness and overall impression, Np also showed high correlation coefficients of 0.31 and 0.45, respectively. This suggests that as an indicator of changes in the diffuse sound field, Np can complement other existing acoustic parameters. In addition, comparing "diffuseness" with other acoustic impression assessment indicators, the relationships with certain physical acoustic parameters such as RT, bass ratio, and Np were at similar levels.

4. Discussion

4.1. *Comparison of computer simulation and scale model*

To compare the results of the computer simulation and scale model

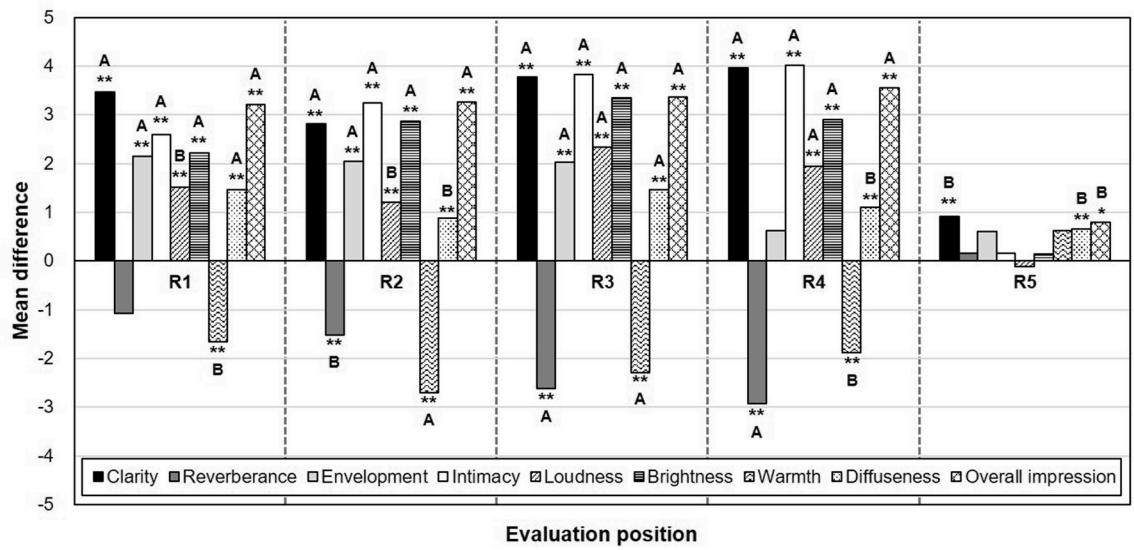


Fig. 7. Mean differences between diffusive and non-diffusive cases at each receiver position (*p < 0.05, **p < 0.01, A: effect > 0.8, B: effect > 0.5).

Table 7
Pearson's correlation coefficients between acoustical parameters and subjective impressions (only cases with p < 0.01 are presented).

	RT	EDT	G	C_{80}	1-$IACC_{E3}$	Brilliance	Bass	Np
Clarity	− .64		− .19	.19	.24	.20	− .41	.47
Reverberance	.38		.22	− .18		-.17	.23	− .32
Envelopment	− .28	.23					− .25	.23
Intimacy	− .52			.21	.16		− .41	.38
Loudness	− .29	.21	.40	.18			− .37	
Brightness	− .50			.19			− .45	.35
Warmth	.30						.20	− .28
Diffuseness	− .47				.18		− .33	.31
Overall impression	− .59				.18		− .39	.45

for the ACI hall, the same five sound collection points were selected under the same wall diffusion state as in Experiment 2 (See the Appendix B). The acoustic parameters (RT_{20}, EDT, G_{mid}, C_{80}, and LF_{E4}) were then calculated and compared. After the application of the diffusers, in the computer simulation model (ACI), RT_{20} (reflective = 2.11 s, diffusive = 2.09 s) changed by −0.02 s, EDT (reflective = 2.54 s, diffusive = 2.48 s) changed by −0.06 s, G_{mid} (reflective = 4.8 dB, diffusive = 4.8 dB) did not change, and C_{80} (reflective = −1.0, diffusive = −0.8) changed by 0.26 dB LF_{E4} (reflective = 0.27, diffusive = 0.27) did not change. All differences were less than the JND. The changes in standard deviation were also very low, less than 0.03 for all parameters. However, when a 1:10 scale model was fabricated and examined before and after wall diffuser installation, the results were considerably different from those obtained via simulation. The values of RT and EDT all positions were greater than the JND, allowing for better observation of the diffusion effect. The simulation and scale models differed in terms of the shape of the aforementioned hall and precision. Furthermore, as the edge diffraction of small facets may be somewhat ignored in the GA software based on Lambert-scattering models [47], these various causes may lead to differences with the actual scale model results. Therefore, for a prediction that is more accurate, it is most desirable to directly create a scale model or perform an evaluation at the actual site. Otherwise, methods will be required to reduce errors, such as implementing a minimum structural size of at least 0.7 m in the simulation while applying actual predictions and corrections [39].

4.2. Influence of sound diffusion on subjective attributes

To investigate the effect of sound diffusion on subjective responses in a music venue, this study performed multiple linear regression analysis using the application of the diffuser as a variable. Eight physical acoustic parameters were used as the independent variables, and Table 8 presents the standardized regression coefficient (β). The overall model was statistically significant as the significance level (p) was less than 0.05; however, the explanatory power differed with the subjective attributes. The indicator that exhibited the largest difference after diffuser application was brightness (β = 0.95), followed by clarity (β = 0.89), intimacy (β = 0.89), and overall impression (β = 0.83). In general, negative relationships were identified only in subjective indicators related to low-frequency reverberation (Reverberance (β = −0.35) and Warmth (β = −0.71)). In addition to diffuser installation, EDT, 1-$IACC_{E3}$, and brilliance had maximum influence on subjective parameters among the physical acoustic parameters. In particular, for the case of diffuseness proposed in this study, the perception rate increased by 0.58 after diffuser attachment, and the contribution of EDT alone was 0.24. This again demonstrates the increase in the density of early reflection generated by diffuse reflection. This is consistent with EDT, one of the major determinants of the diffuse sound field preference in previous research [22]. In addition, compared to the study of Shtrepi [28] that found no relationship between physical and subjective indicators such as reverberance, sound level, coloration, spaciousness, and source localization, the diffuseness indicator proposed in this study can be considered a more advanced indicator. However, with a low explanatory power of 0.30 and no relationship identified with other indicators, it is

Table 8
Standardized regression coefficients (β) obtained from the multiple linear regression analysis of subjective attributes using objective parameters (Only cases with $p < 0.05$ are presented. Underlined: $p < 0.05$, Bold: $p < 0.01$).

	R^2	Diffuser	RT	EDT	G	C_{80}	1-$IACC_{E3}$	Brilliance	Bass	Np
Clarity	**.75**	**.89**	**.42**	**.37**			**-.28**	**.73**		**.30**
Reverberance	**.23**	-.35		-.18			**.26**	-.38		
Envelopment	**.20**	**.45**		**.30**						
Intimacy	**.47**	**.89**	**.61**	**.40**	.19		**-.36**	**.81**	-.16	**.40**
Loudness	**.42**	**.56**		**.33**	**.54**		**-.21**	**.48**		
Brightness	**.49**	**.95**	**.61**	**.43**	**.25**		**-.36**	**.75**	-.16	**.34**
Warmth	**.20**	**-.71**	-.44	-.28			**.36**	**-.44**		
Diffuseness	**.30**	**.58**		.24						
Overall impression	**.69**	**.83**	**.51**	.36	.17		**-.27**	**.72**		**.41**

necessary to consider other factors related to diffusion perception in addition to the four evaluation factors of diffusion examined in this study.

4.3. Limitations of this study

This study has the following limitations. It proposed the concept of "diffuseness" to evaluate the perception of diffusion in music venues. To evaluate diffuseness, researchers arbitrarily selected four items based on subjective judgment, but even if evaluated by a group of experts, it is very difficult to distinguish each item by ear. Therefore, in future research, the optimal attributes for evaluating diffusion should be investigated in advance through pilot tests to propose a diffusion perception indicator that can be more clearly interpreted than the "diffuseness" indicator proposed in this study. Furthermore, as verified by Shtrepi et al. [28], more realistic geometrical modeling of the diffuser shape on the wall, rather than simply applying the scattering coefficient to a flat surface on the simulation model, will allow for a more realistic verification of auditory diffuse reflections after diffuser attachment and could achieve better agreement with the measurements. For the scale model, only one model and five evaluation points in the auditorium were selected for subjective evaluation, which may be insufficient to generalize the subjective perceptive characteristics of diffusion. Nevertheless, such an approach is meaningful in that the comparative evaluation of one hall clearly showed the effects of diffusion.

5. Conclusions

In this study, the changes in the sound field before and after the application of diffusive design to a musical venue were investigated using computer simulation and a scale model, in terms of objective and subjective aspects. In addition, nine subjective perceptive characteristics were investigated using audible sound sources in the scale model. Although the differences in the objective indicators, except for reverberation (RT and EDT), were not clear, the subjects were well aware of the changes in the diffuse sound field owing to the diffuser attached to the sidewall. Consequently, in the scale model, the effect of the vertical diffuser on the early reflections of the audience around the stage was clear. Moreover, the reverberance sound field formed by the reflections from the entire surface of the space strongly influenced the rear seats. In this regard, this study presents the new finding that the installation of diffusers in a musical venue not only mitigates the acoustic problems of specular reflection but also enhances the early and late acoustic characteristics in the time domain of the sound occurring in the space. For the rear seats, the acoustic improvement due to the presence of the diffuser resulted in a decrease in reverberation and an increase in clarity. Thus, the changes in clarity and reverberation can be seen as a transitional effect of diffuser utilization, which can lead to changes in the sound field of the concert hall. This indicates that the effect of the diffuser is perceived through changes in the temporal distribution of the side reflection energy. However, additional research is required to determine whether this perception of diffusion is positive or negative.

The novel contributions of this study are as follows. (1) With "diffuseness" newly defined as a subjective characteristic of diffusion and the four psychoacoustic expressions ("density of reflections," "smoothness of decay," "smoothness of reflection," and "isotropic directivity"), the subjective preference for the diffuse sound field can be evaluated. (2) An increase in the number of early reflections due to the diffusion effect influences the subjective evaluation. Therefore, Np can also serve as a useful supplement, in addition to the commonly used physical parameters, such as RT and bass ratio, to explain the results of subjective evaluations of the diffuse sound field. (3) The use of a diffuser results in a decrease in the RT, owing to an increase in sound absorption. Ultimately, when a diffuser is installed on a wall near the sound source on the stage, it decreases the intensity of the early reflections but increases the density of the early reflections produced in various directions. This results in increased EDT(s), and even with the decreased G, the audible loudness and diffuseness increase, thus, meeting the subjects' preference. The venue tends to be more affected by the diffuse nature of the reflections. Therefore, it is effective to consider the decay curve pattern of the early reflections and the change in the early reflections with diffusion in the time domain of the impulse response in the early stage of designing the venue. Future studies should generalize the impact of the diffuser on subjective cognition by evaluating various performance halls. Moreover, such studies should consider perceptive changes that occur in spaces other than a musical venue. In addition to linguistic performances, such as vocal music, the influence of diffusers should also be studied with regard to other forms of performances, such as symphonies and concertos. As demonstrated in this study, changes higher than the JND in audience acoustics after diffuser application were not found by the computer simulation. However, the measurements in the scaled model showed differences higher than the JND in the reverberation-related indices (RT and EDT). This suggests that the GA computer simulation still has technical limitations in evaluating the diffuse sound field. To overcome this limitation, additional algorithm research is required. Therefore, computer simulations and scaled modeling should be considered together for a more accurate acoustic prediction in the early stage of the design of a music venue. Furthermore, even though the changes in objective indices other than reverberation after diffuser attachment were not higher than the JND, noticeable auditory differences were found in subjective evaluations. Therefore, for a more in-depth analysis considering diffusion and the like, auditory evaluations through auralization are necessary. The computer simulation results and the new diffusion indicator, "diffuseness," identified in this study will be useful for music venue designers and acoustic engineers.

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Declaration of competing interest

None.

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Appendix A. Summary of room acoustical parameters according to different scattering coefficients for 12 concert hall models

Scattering coefficient			RT_{20} [s]			EDT [s]			G_{mid} [dB]			C_{80} [dB]			LF_{E4}		
			0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ	0.05	0.7	Δ
Shoebox	BO	*Max*	3.28	2.44	−0.84	2.94	2.42	−0.52	9.5	9.5	0	4.6	4.2	−0.4	0.37	0.37	0
		Min	2.13	1.82	−0.31	1.68	1.71	0.03	2.0	1.3	−0.7	−4.9	−4.6	0.3	0.11	0.10	−0.01
	VM	*Max*	2.91	2.82	−0.09	2.85	2.95	0.1	14.7	14.7	0	7.7	7.8	0.1	0.31	0.31	0
		Min	2.64	2.58	−0.06	2.21	2.28	0.07	2.5	1.8	−0.7	−3.2	−3.5	−0.3	0.02	0.02	0
	AM	*Max*	2.34	2.36	0.02	2.77	2.62	−0.15	9.0	8.9	−0.1	3.0	3.2	0.2	0.27	0.27	0
		Min	2.24	2.23	−0.01	2.26	2.20	−0.06	3.5	3.1	−0.4	−2.7	−2.1	0.6	0.04	0.04	0
	LF	*Max*	1.65	1.63	−0.02	1.89	1.83	−0.06	11.0	11.0	0	10.0	10.3	0.3	0.31	0.31	0
		Min	1.48	1.47	−0.01	0.64	0.61	−0.03	−0.7	−1.3	−0.6	−4.0	−3.7	0.3	0.03	0.03	0
Fan	SA	*Max*	1.98	2.03	0.05	2.39	2.43	0.04	10.7	10.7	0	6.2	6.4	0.2	0.38	0.32	−0.06
		Min	1.78	1.86	0.08	1.46	1.35	−0.11	3.0	1.4	−1.6	−4.3	−4.4	−0.1	0.02	0.01	−0.01
	GO	*Max*	1.69	1.88	0.19	1.84	2.08	0.24	10.7	10.8	0.1	6.6	6.0	−0.6	0.39	0.32	−0.07
		Min	1.53	1.47	−0.06	1.45	1.65	0.2	3.1	3.1	0	−2.6	−4.0	−1.4	0.03	0.03	0
	MU	*Max*	2.75	2.33	−0.42	3.42	4.16	0.74	9.2	9.1	−0.1	4.4	4.7	0.3	0.39	0.43	0.04
		Min	2.10	2.00	−0.1	1.81	1.68	−0.13	−2.4	−3.4	−1	−9.8	−9.8	0	0.03	0.03	0
	ACI	*Max*	2.18	2.2	0.02	2.65	2.62	−0.03	6.53	6.87	0.34	−0.13	0.87	1	0.33	0.34	0.01
		Min	2.04	2.06	0.02	2.21	2.08	−0.13	1.7	1.5	−0.2	−3.13	−3.37	−0.24	0.17	0.18	0.01
Other	EB	*Max*	1.75	1.77	0.02	2.20	2.20	0	10.6	10.7	0.1	5.4	5.6	0.2	0.47	0.51	0.04
		Min	1.50	1.55	0.05	1.55	1.33	−0.22	0.2	0.2	0	−3.7	−3.1	0.6	0.03	0.03	0
	LB	*Max*	2.23	2.11	−0.12	2.28	2.22	−0.06	6.9	6.8	−0.1	1.83	1.7	−0.13	0.33	0.2	−0.13
		Min	2.05	1.89	−0.16	1.88	1.91	0.03	2.3	0.6	−1.7	−5.6	−4.9	0.7	0.03	0.03	0
	ST	*Max*	2.41	3.00	0.59	2.43	2.43	0	11.4	11.3	−0.1	7.2	7.5	0.3	0.40	0.37	−0.03
		Min	1.77	1.66	−0.11	1.51	1.40	−0.11	0.1	0.0	−0.1	−4.6	−4.3	0.3	0.01	0.01	0
	CW	*Max*	2.26	2.39	0.13	2.41	2.28	−0.13	9.1	9.2	0.1	6.3	6.2	−0.1	0.35	0.37	0.02
		Min	1.72	1.95	0.23	1.29	1.51	0.22	−2.4	−3	−0.6	−5.7	−5.4	0.3	0.01	0.02	
Average		*Max*	2.29	2.25	−0.04	2.51	2.52	0.01	10.0	10.0	0.0	5.3	5.4	0.1	0.36	0.34	−0.02
		Min	1.92	1.88	−0.04	1.66	1.64	−0.02	1.1	0.4	−0.6	−4.5	−4.4	0.1	0.04	0.04	0.00

Appendix B. Comparison of acoustic parameters simulated in diffusive and non-diffusive cases for ACI. (bold numbers indicate delta values higher than the JND)

	Diffuser	RT_{20}[s]	EDT[s]	C_{80}[dB]	G_{mid}[dB]	LF
R1	Without	2.09	2.54	1.4	6.7	0.34
	With	2.15	2.48	1.7	6.8	0.33
	Δ	0.06	−0.06	0.3	0.1	−0.01
R2	Without	2.09	2.49	0.1	6.6	0.28
	With	2.13	2.38	0.7	6.8	0.28
	Δ	0.04	−0.11	0.6	0.2	0
R3	Without	2.07	2.63	−1.8	5	0.3
	With	2.06	2.59	−1.5	4.9	0.29
	Δ	−0.01	−0.04	0.3	−0.1	−0.01
R4	Without	2.11	2.57	−2.0	3.1	0.26
	With	2.06	2.52	−1.7	3.1	0.27
	Δ	−0.05	−0.05	0.3	0	0.01
R5	Without	2.17	2.46	−2.6	2.5	0.18
	With	2.04	2.45	−3.1	2.5	0.18
	Δ	**−0.13**	−0.01	−0.5	0	0
Ave	Without	2.11	2.54	−1.0	4.8	0.27
	With	2.09	2.48	−0.8	4.8	0.27
	Δ	−0.02	−0.06	0.2	0	0
Stdev	Without	0.04	0.07	1.7	2	0.06
	With	0.05	0.09	1.9	2	0.06
	Δ	0.01	0.02	0.3	0	0

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